Richard Cao

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Education

High School: Beijing National Day School (BNDS) **Class of 2026**
GPA 4.24/4.3 | SAT 1570 (770 Reading & Writing | 800 Math) | AP Exam Scores: eleven 5s, two 4s.
Advanced Coursework: Electrodynamics, Introductory Quantum Mechanics, and Introductory Analytic Mechanics
UC Berkeley: Double Major in Physics and Mathematics **Class of 2030**

Honors & Awards

- High School Mathematics Contest in Modeling (HiMCM) Finalist (Top 5–7%) 2024
- British Physics Olympiad (BPhO) Global Gold (Top 13%) 2024
- Physicists' Tournaments (phenomena modeling) – 1st & 2nd Prizes (Beijing Normal University Physicists' Tournament; provincial/national contests) 2024–2025
- American Association of Physics Teachers HS Physics Photo Contest, Top 100 Finalist 2025
- 1st Prize Formative Assessment (highest school award for grades) 2023–2025

Research & Technical Experience

Core Member, BNDS Physicists' Tournament Team 2023 -2025

- Investigated open-ended physics problems spanning symmetry breaking, non-linear dynamics, and beam dynamics (Spaghetti Accelerator; Rayleigh–Bénard convection; Lato-Lato).

Optics Researcher (Youth Talent Program, selective national research) Dec 2023 – Oct 2024

- Guided by Peking University professor Sun Dong, built and stabilized an ultrafast confocal microscopy module. Studied Köhler illumination, photoluminescence mechanisms, and data analysis. Learned how to collimate laser beams by adjusting mirrors.

Quarterback Throwing Motion Analyzer 2024 – 2025

- Built software using YOLO v8 (a computer-vision model) to visualize quarterback throwing techniques by comparing one's motion to that of elite NFL quarterbacks; achieved by triangulating coordinates of two different camera angles.

Quantitative Stock Prediction via Ising Model 2026 – Present

- Attempted to apply 2D Ising model principles of ferromagnetic spin orientation to modeling buy/sell behavior of stockholders, trying to mirror how individuals base decisions on information in their radius of knowledge instead of on the entire landscape.

Leadership

President & Offensive Captain of BNDS Dragons Flag Football 2024 – 2026

- Played as the quarterback; organized 5 hours/week practices. Mentored rookie QBs. Helped the team achieve an undefeated streak of 10 games in 2024-2025.
- Co-founded the Beijing High School Football League, a student-led flag football league, with 8 other schools.

Personal Side Quests

"Meet Physics" channel 2025 - 2026

- Founded my "Meet Physics" channel with 4 friends I met at the Physics Tournament; received 1.4M+ views; got to appreciate the power of asking good (interesting physics) questions.

Film photography 2025 – present

- Shot ∼20 rolls; developed 1 on my own. I love film photography. It slows me down as I revisit old spots and explore new places. Though it is more expensive in the US.